

Joint Semi-Symmetric Tensor PCA for Integrating Multi-modal Populations of Networks

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Joint work with Jiaming Liu, Zhengwu Zhang, Genevera I. Allen

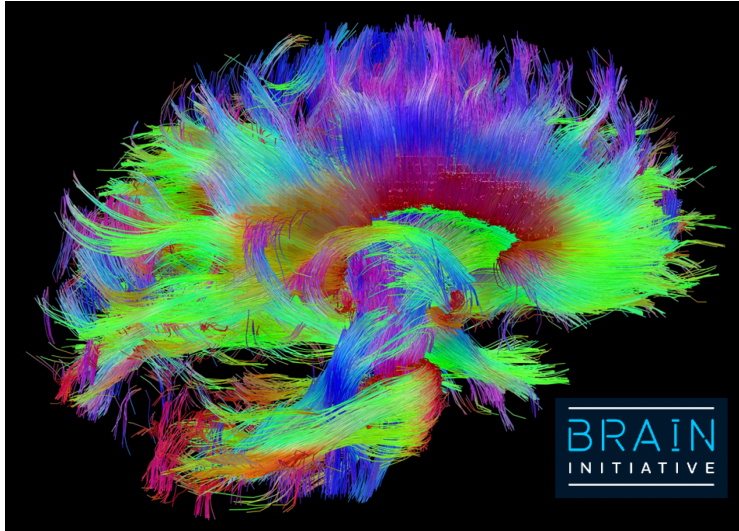
1. Motivation
2. JisstPCA Model & Algorithm
3. Theoretical Properties
4. Empirical Results

Motivation

Motivation

Integrating Brain Connectomes

Large-Scale Neuroimaging Studies



Human Connectome Project (HCP):

- Over 1200 healthy adults.
- Multi-modal brain imaging: **Diffusion MRI (dMRI)**, **Functional MRI (fMRI)**, structural MRI, Magnetoencephalography (MEG) and electroencephalography (EEG).
- Genomic data.
- Demographic, clinical and behavioral data.
- Cognitive testing data.

(Van Essen et al., 2012)

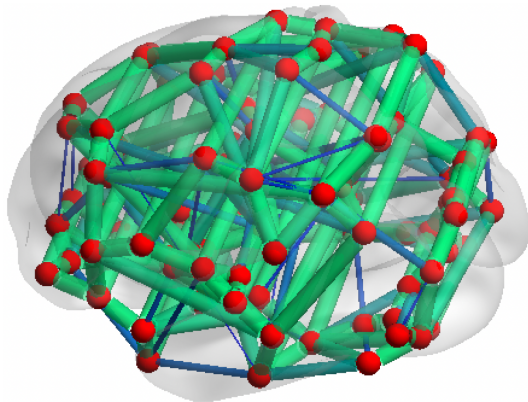
Other Studies: UK Biobank, ABIDE, ADNI, etc.

Motivating Question

How do we integrate all of this multi-modal & multi-subject data to better understand brain connectomes?

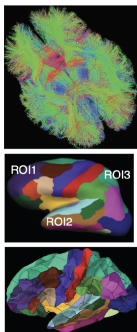
Brain Connectomes

Brain connectomes use networks to understand how the brain is structurally and functionally connected.



Structural Connectome (from dMRI data):

Tractography
+
Parcellation



finer resolution

Complex

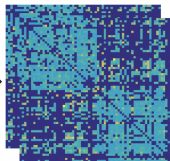
Simple

Streamline level

	ROI1	ROI2	ROI3
ROI1			
ROI2			
ROI3			

Streamline Connectivity
Cell Matrix (SCCM)

Weighted network level



Scalar Matrices

Binary network level

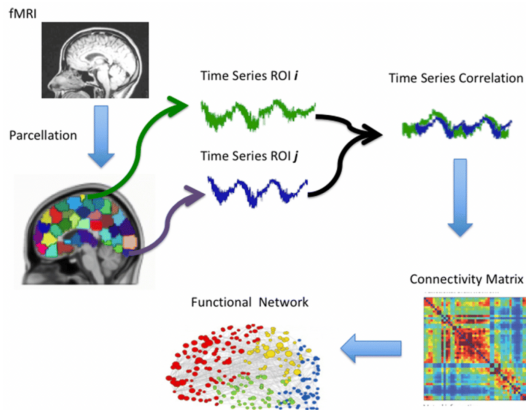


Binary Matrix

Current literature

Z. Zhang, M. Descoteaux, A. Srivastava, D. Dunson, H. Zhu, et al.
Mapping Population-based Structural Connectomes, *Neuroimaging*

Functional Connectome (from fMRI data)



(Undirected / Static Network; Milano et al., 2019)

Challenges with Analyzing Brain Connectomes in HCP Data:

- Multi-modal networks.
(Han et al., 2015; D'Angelo et al., 2019; Gao et al., 2022; Sui et al., 2012; Yao et al., 2015; Cole et al., 2021)
- Multi-subject (population) networks.
(Paul and Chen, 2020; Weylandt and Michailidis, 2022; Zhang et al., 2019; Winter et al., 2020; Wang et al., 2019; Pavlović et al., 2020)

No existing approaches for analyzing both aspects!

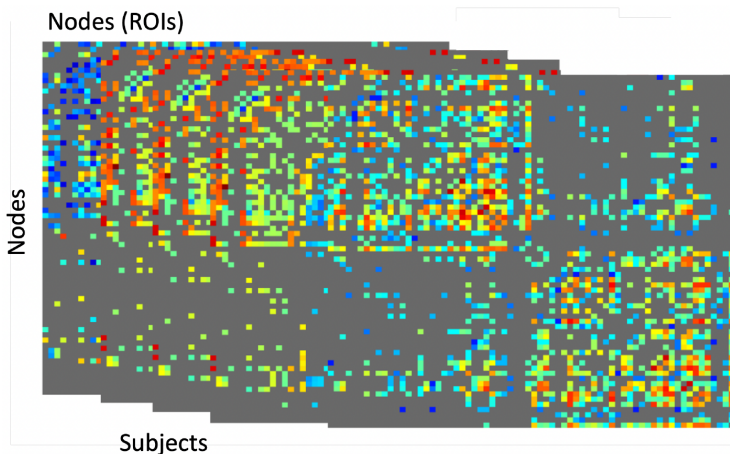
Our Goal

Our goal is to develop an interpretable dimension reduction model and approach (with rigorous statistical guarantees) for uncovering patterns in multi-modal populations of networks.

Brain Connectomes as Tensors

Organize a population of n (undirected) networks as a semi-symmetric tensor:

- Stacked weighted adjacency or Laplacian matrices.



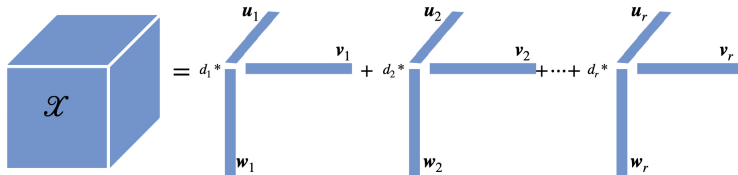
Motivation

Background: Tensor PCA

Background: Tensor Decompositions

No direct extension of SVD for tensors!

Candecomp / Parafac (CP) Model:

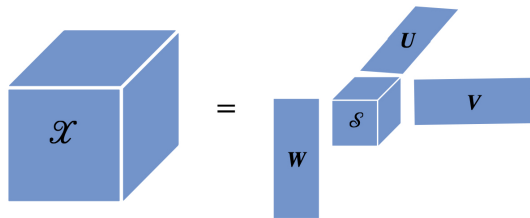


Main Algorithms: Alternating Least Squares (CP-ALS) & Power Method.
(Carroll and Chang, 1970; Anandkumar et al., 2014; Han and Zhang, 2022)

Background: Tensor Decompositions

No direct extension of SVD for tensors!

Tucker Model:



Main Algorithms: Higher-Order SVD (HOSVD) & Higher-Order Orthogonal Iteration (HOOI).

(De Lathauwer et al., 2000a; Zhang and Xia, 2018; Luo et al., 2021)

No direct extension of SVD for tensors!

Statistical Challenges:

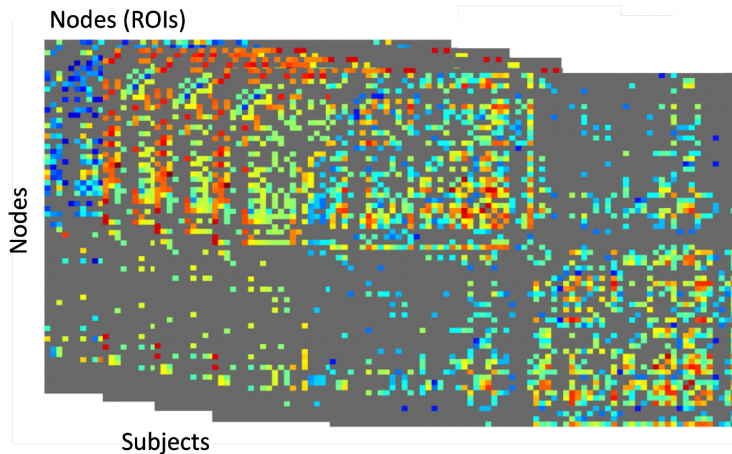
- Non-convex & dependent on initialization.
- Algorithmic convergence not guaranteed.
- What decomposition is appropriate for the desired application?

Statistical Analysis of Tensor PCA:

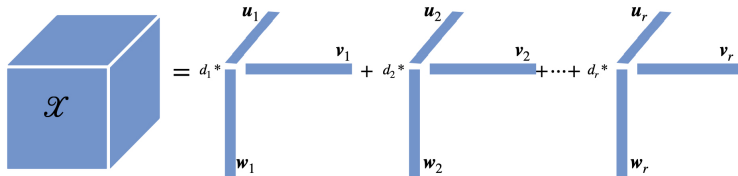
(Richard & Montanari, 2014; Arous et al., 2019; Auddy & Yuan, 2022; Auddy & Yuan, 2023; Zhang & Xia, 2018; Anandkumar et al., 2014; Luo et al., 2021; Janzamin et al., 2019; Ouyang & Yuan, 2023).

Prior Work: Semi-Symmetric Tensor PCA

Motivation: Population analysis of Structural Connectomes.



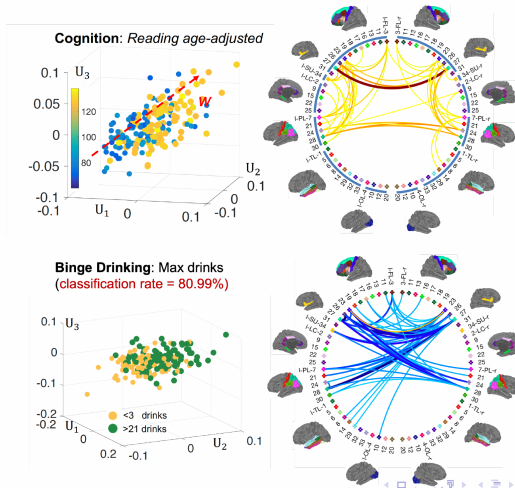
Statistical Modeling: Semi-Symmetric Tensor PCA (CP Model).
(Zhang et al., 2019)



- Extended model and theoretical analysis by (Weylandt & Michailidis, 2022)

Prior Work: Semi-Symmetric Tensor PCA

Results Preview: Principal Brain Network (SC) and Association with Traits



Question:

Can we extend this semi-symmetric tensor model for multi-modal networks (e.g. integrating Structural & Functional Connectomes)?

JisstPCA Model & Algorithm

JisstPCA Model & Algorithm

JisstPCA Model

JisstPCA Model:

$$\mathcal{X} = \sum_{k=1}^K d_{x,k}^* \cdot V_k^* V_k^{*\top} \circ u_k^* + \mathcal{E}_x, \quad \mathcal{Y} = \sum_{k=1}^K d_{y,k}^* \cdot W_k^* W_k^{*\top} \circ u_k^* + \mathcal{E}_y.$$

- $\mathcal{X} \in \mathbb{R}^{p \times p \times N}$, $\mathcal{Y} \in \mathbb{R}^{q \times q \times N}$. E.g., $\mathcal{X}_{:, :, i} \in \mathbb{R}^{p \times p}$ is the functional connectome associated with subject i ; $\mathcal{Y}_{:, :, i} \in \mathbb{R}^{q \times q}$ is the structural connectome also for subject i .

JisstPCA Model:

$$\mathcal{X} = \sum_{k=1}^K d_{x,k}^* \cdot V_k^* V_k^{*\top} \circ u_k^* + \mathcal{E}_x, \quad \mathcal{Y} = \sum_{k=1}^K d_{y,k}^* \cdot W_k^* W_k^{*\top} \circ u_k^* + \mathcal{E}_y.$$

- K -factor low-rank tensor + noise ($\mathcal{E}_x \in \mathbb{R}^{p \times p \times N}$, $\mathcal{E}_y \in \mathbb{R}^{q \times q \times N}$ are random zero-mean semi-symmetric noise tensors).

Joint, Integrative Semi-Symmetric Tensor PCA (JisstPCA)

JisstPCA Model:

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- K -factor low-rank tensor + noise ($\mathcal{E}_x \in \mathbb{R}^{p \times p \times N}$, $\mathcal{E}_y \in \mathbb{R}^{q \times q \times N}$ are random zero-mean semi-symmetric noise tensors).
- $V_k^* \in \mathcal{O}_{p, r_{x,k}}$, $W_k^* \in \mathcal{O}_{q, r_{y,k}}$ are orthogonal and the k th principal network factors.
- $(V_k^* V_k^{*\top}, W_k^* W_k^{*\top})$ are network prototypes.
- $u_k^* \in \mathbb{S}^{N-1}$ is the joint population factor.
- $d_{x,k}^*, d_{y,k}^*$ are eigenvalues of the k th network prototype, with multiplicity $r_{x,k}$ and $r_{y,k}$.

Joint, Integrative Semi-Symmetric Tensor PCA (JisstPCA)

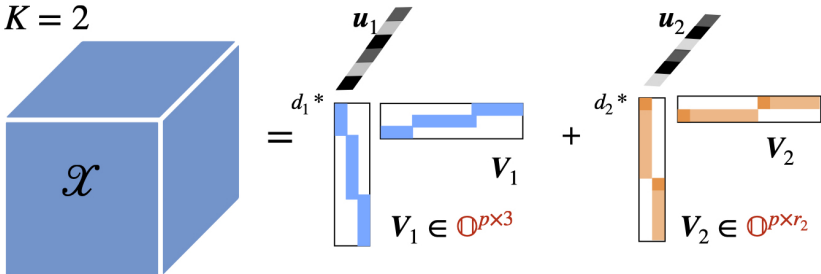
The diagram illustrates the JisstPCA decomposition of a 3D tensor $\mathcal{X}$. On the left, a blue cube represents the tensor $\mathcal{X}$. To its right, an equals sign indicates the decomposition into a sum of rank-1 components. The first component consists of a vertical blue bar representing a vector u_1 , a horizontal blue bar representing a matrix V_1 , and a scalar d_1^* positioned near the top of the bar. Below this component, the text $V_1 \in \mathbb{O}^{p \times r_1}$ is shown. This is followed by a plus sign and an ellipsis, and then the K -th component, which similarly consists of a vertical bar for u_K , a horizontal bar for V_K , and a scalar d_K^* . Below this component, the text $V_K \in \mathbb{O}^{p \times r_K}$ is shown.

$$\mathcal{X} = d_1^* u_1 V_1 + \dots + d_K^* u_K V_K$$

$V_1 \in \mathbb{O}^{p \times r_1}$ $V_K \in \mathbb{O}^{p \times r_K}$

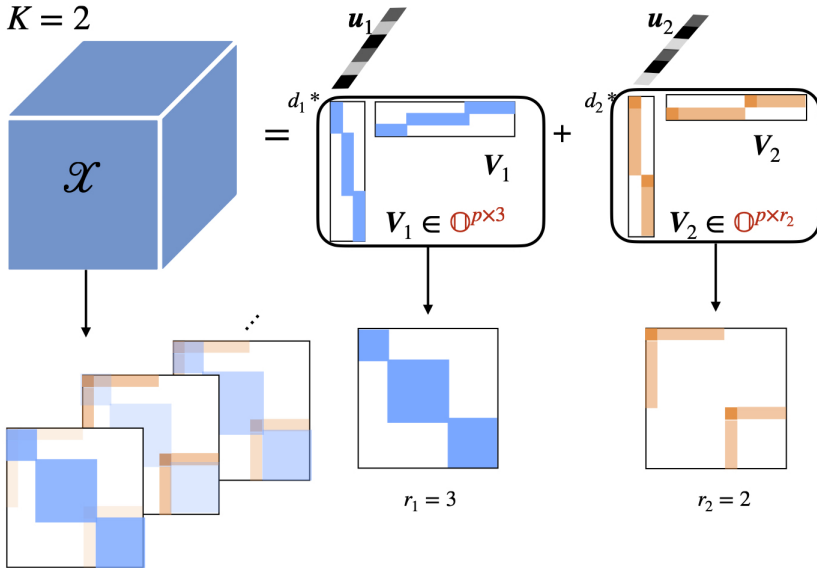
Joint, Integrative Semi-Symmetric Tensor PCA (JisstPCA)

Example: $K = 2$



Joint, Integrative Semi-Symmetric Tensor PCA (JisstPCA)

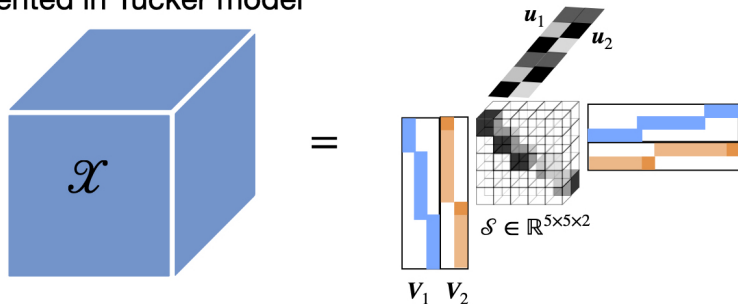
Example: $K = 2$



Relation to Existing Models

Not quite a Tucker or CP model:

Represented in Tucker model



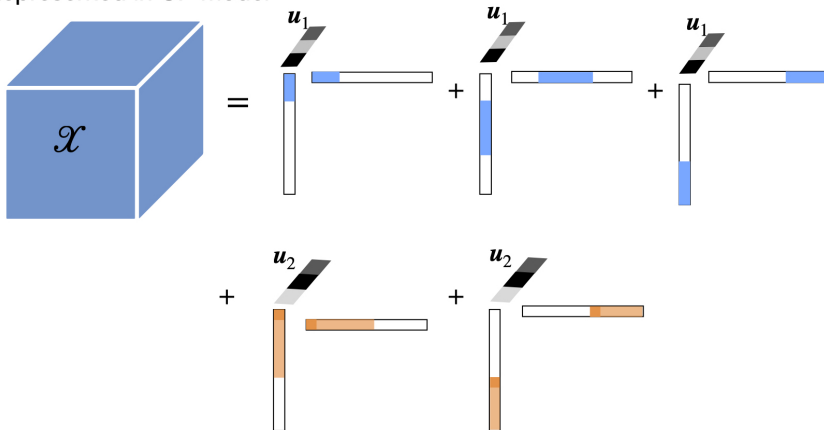
subject to orthogonality constraint!

But, better for representing multi-modal & multi-subject networks!

Relation to Existing Models

Not quite a Tucker or CP model:

Represented in CP model



Repeated u_k is not allowed by existing CP methods!

Other Tensor Network Models:

- Semi-symmetric tensor PCA (Weylandt and Michailidis, 2022).
- Semi-symmetric CP decomposition (Zhang et al., 2019; Winter et al., 2020; Wang et al., 2014).

Data Integration Models:

- Matrix integration (Lock et al., 2013; Tang & Allen, 2021; Abdi et al., 2013; Acar et al., 2011; Risk & Gaynanova, 2021).
- Tensor integration via CP decomposition (Acar et al., 2011, 2014; Wu et al., 2018; Schenker et al., 2020; Lu et al., 2020; Farias et al., 2016; Fu et al., 2015).

Other Low-Rank Network Models:

- Latent Factor Models (Paul and Chen, 2020; MacDonald et al., 2022; Zhang et al., 2020).

JisstPCA Model & Algorithm

JisstPCA Algorithm

JisstPCA Problem:

$$\arg \min_{\{u_k, V_k, W_k, d_{x,k}, d_{y,k}\}} \lambda \left\| \mathcal{X} - \sum_{k=1}^K d_{x,k} \cdot V_k V_k^\top \circ u_k \right\|_F^2 + (1-\lambda) \left\| \mathcal{Y} - \sum_{k=1}^K d_{y,k} \cdot W_k W_k^\top \circ u_k \right\|_F^2$$

s.t. $V_k \in \mathcal{O}_{p, r_{x,k}}, \quad W_k \in \mathcal{O}_{q, r_{y,k}}, \quad u_k \in \mathbb{S}^{N-1}.$

$\lambda \in [0, 1]$ is a hyperparameter to account for data on different scales.

Challenge: Non-Convex!

Single-factor JisstPCA

- **Input:** $\mathcal{X}$, $\mathcal{Y}$, r_x , r_y , λ , $t_{\max}$, & initial $u^{(0)}$.
- **repeat until** $t = t_{\max}$ **or convergence:**

$$V^{(t+1)} = \text{Leading } r_x \text{ singular vectors of } \mathcal{X} \times_3 u^{(t)}$$

$$W^{(t+1)} = \text{Leading } r_y \text{ singular vectors of } \mathcal{Y} \times_3 u^{(t)}$$

$$u^{(t+1)} = \frac{\lambda[\mathcal{X}; V^{(t+1)}] + (1 - \lambda)[\mathcal{Y}; W^{(t+1)}]}{\|\lambda[\mathcal{X}; V^{(t+1)}] + (1 - \lambda)[\mathcal{Y}; W^{(t+1)}]\|_2}$$

$$t = t + 1$$

- **return** $\hat{u} = u^{(t)}$, $\hat{V} = V^{(t)}$, $\hat{W} = W^{(t)}$; $\hat{d}_x = \langle X, \hat{V} \hat{V}^\top \circ \hat{u} \rangle / r_x$, $\hat{d}_y = \langle Y, \hat{W} \hat{W}^\top \circ \hat{u} \rangle / r_y$; $\hat{\mathcal{X}} = \hat{d}_x \cdot \hat{V} \hat{V}^\top \circ \hat{u}$, $\hat{\mathcal{Y}} = \hat{d}_y \cdot \hat{W} \hat{W}^\top \circ \hat{u}$.

Single-factor JisstPCA

- **Input:** $\mathcal{X}$, $\mathcal{Y}$, r_x , r_y , λ , $t_{\max}$, & initial $u^{(0)}$.
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$$u^{(t+1)} = \frac{\lambda[\mathcal{X}; V^{(t+1)}] + (1 - \lambda)[\mathcal{Y}; W^{(t+1)}]}{\|\lambda[\mathcal{X}; V^{(t+1)}] + (1 - \lambda)[\mathcal{Y}; W^{(t+1)}]\|_2}$$

$t = t + 1$

- Power Method.
- Dependent on initialization; Convergence not guaranteed.

Generalized JisstPCA:

$$\mathcal{X} = \sum_{k=1}^K V_k^* D_{x,k}^* V_k^{*\top} \circ u_k^* + E_x, \quad \mathcal{Y} = \sum_{k=1}^K W_k^* D_{y,k}^* W_k^{*\top} \circ u_k^* + E_y.$$

- $D_{x,k}^* \in \mathbb{R}^{r_{x,k} \times r_{x,k}}$ and $D_{y,k}^* \in \mathbb{R}^{r_{y,k} \times r_{y,k}}$ are diagonal matrices.
- Each network prototype does not share a multiplicity of eigenvalues $\Rightarrow$ more flexible network prototype representation.
- **Algorithm:** Extension of JisstPCA algorithm.

Other Extensions:

- Matrix + semi-symmetric tensor integration.
- Integration of more than two tensors.

Scaling (λ):

Suggested default:

$$\lambda = \frac{\|X^*\|_F}{\|X^*\|_F + \|Y^*\|_F}.$$

Number of layers (K):

Cumulative proportion of variance explained (Allen, 2012).

Ranks (r_x and r_y):

BIC + Deflation approach (Allen, 2012; Zhou et al., 2022; Hu et al., 2022):

- Choose ranks for each factor via BIC and then deflate.

Theoretical Properties

Assumptions:

- A3. **(Sub-Gaussian Noise)** Let $(\mathcal{E}_x)_{:, :, k}$ and $(\mathcal{E}_y)_{:, :, k}$ have independent, zero-mean, sub-Gaussian- σ entries subject to symmetry constraints.
- A4. **(SNR Condition)** Let

$$d_\lambda = \sqrt{\lambda r_x d_x^{*2} + (1 - \lambda) r_y d_y^{*2}},$$

and suppose

$$d_\lambda \geq C\sigma \left(N^{1/4} (\sqrt{p} + \sqrt{q}) + \sqrt{N} \right) \sqrt{\log N}$$

for some constant $C > 0$. In addition, suppose $d_x^* \geq C\sigma(\sqrt{N} + \sqrt{p})$, $d_y^* \geq C\sigma(\sqrt{N} + \sqrt{q})$.

Main Theorem

Assuming A3 & A4, the JisstPCA Algorithm ($K = 1$) with spectral initialization satisfies the following w.h.p.:

$$|\sin \theta(u, u^{(1)})| \leq \frac{C\sigma(\lambda r_x(\sqrt{p} + \sqrt{N}) + (1 - \lambda)r_y(\sqrt{q} + \sqrt{N}))}{\lambda r_x d_x^* + (1 - \lambda)r_y d_y^*},$$

$$\|\sin \Theta(V, V^{(2)})\|_{op} \leq \frac{C\sigma(\sqrt{p} + \sqrt{N})}{d_x^*}, \quad \|\sin \Theta(W, W^{(2)})\|_{op} \leq \frac{C\sigma(\sqrt{q} + \sqrt{N})}{d_y^*}.$$

- Improves upon prior work: When $r_x = r_y = r$, $d_x^* = d_y^* = d$, $p = q$, error scales as

$$\frac{\sigma(\sqrt{p} + \sqrt{N})}{d},$$

which improves upon result from (Weylandt & Michailidis, 2022) on non-integrated data

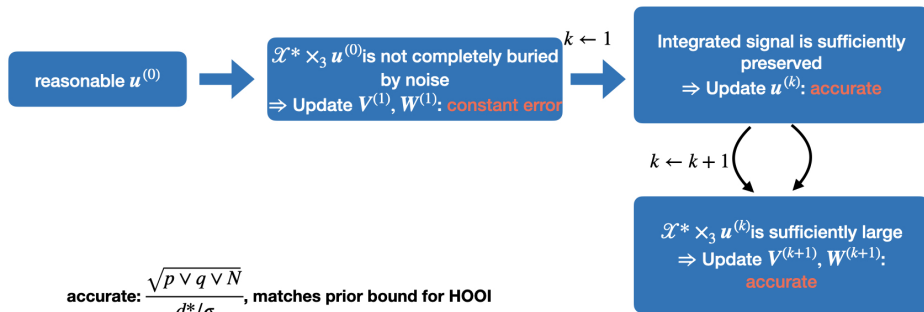
$$\frac{\sigma r \sqrt{pN}}{d}.$$

- Analogous to rates for HOOI:

$$\frac{\sigma \sqrt{p}}{d} \quad \text{for HOOI with Tucker model (Zhang \& Xia, 2018; Luo et al., 2021).}$$

Statistical Convergence Guarantee

- Proof sketch and intuition behind one-step convergence



Assumptions:

A5. **(SNR for Joint Factors)** The integrated signal satisfies:

$$\lambda \|D_x^*\|_F^2 + (1 - \lambda) \|D_y^*\|_F^2 \geq C\sigma^2 \left(\frac{p}{N(p^2 + q^2)} + N \right) \log N,$$

and the individual signals satisfy

$$\|D_x^*\|_F^2 \geq C\sigma^2 r_x(p + N), \quad \|D_y^*\|_F^2 \geq C\sigma^2 r_y(q + N).$$

A6. **(SNR for Network Factors)**

$$\sigma_{r_x}^2(D_x^*) \geq C\sigma^2(p + N), \quad \sigma_{r_y}^2(D_y^*) \geq C\sigma^2(q + N).$$

G-JiistPCA Convergence

Assuming A3 & A5, the G-JiistPCA Algorithm ($K = 1$) with spectral initialization satisfies the following w.h.p.:

$$|\sin \theta(u^*, u^{(k)})| \leq \frac{C\sigma\sqrt{r_x(p+N)}}{\|D_x^*\|_F} \vee \frac{C\sigma\sqrt{r_y(q+N)}}{\|D_y^*\|_F}.$$

Also assuming A6:

$$\left\| \sin \Theta(V^*, V^{(2)}) \right\|_{op} \leq \frac{C\sigma\sqrt{p+N}}{\sigma_{r_x}(D_x^*)}, \quad \left\| \sin \Theta(W^*, W^{(2)}) \right\|_{op} \leq \frac{C\sigma\sqrt{q+N}}{\sigma_{r_y}(D_y^*)}.$$

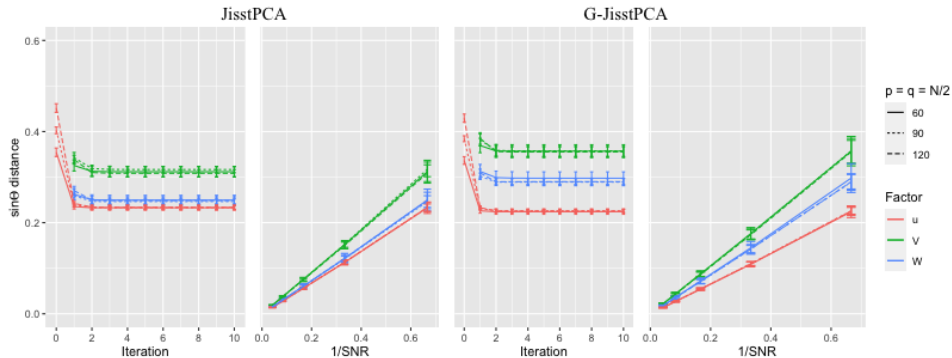
No minimum eigenvalue condition: robust to rank misspecification!

Empirical Results

Empirical Results

Simulation Studies

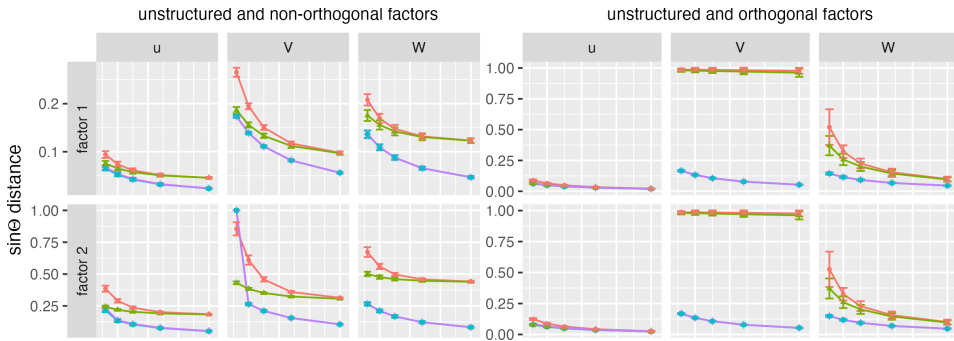
Theory Validation



- Establishes one-step statistical convergence.
- Shows rate is proportional to $1/\text{SNR}$ (σ/d_λ).

Random, unstructured factors (JisstPCA Model)

Simulation 1: Random, unstructured factors (JisstPCA Model)



- $K = 2$ factors, ranks $r_x = r_y = (3, 2)^\top$, $p = 150$, $q = 50$, and $N = 50$.
- Compared to iHOOI & iHOSVD (integrative extensions of Tucker models).

Simulation 2: Structured network factors (JisstPCA Model)



Simulation 3: Network Model (Stochastic Block Model)

Simulation Set-up:

- N samples divided randomly into two clusters ($K = 2$), with probability 0.75 and 0.25, respectively.
- Each cluster is associated with a pair of SBMs: $(M_{k,x}, M_{k,y})$.
- For any sample i from cluster k , its associated networks $X_{:, :, i}$ and $Y_{:, :, i}$ are binary adjacency matrices sampled from $M_{k,x}$ and $M_{k,y}$, respectively.
- $M_{1,x}$ and $M_{1,y}$ are three-block SBMs and $M_{2,x}$ and $M_{2,y}$ are two-block SBMs, with within-block probabilities in $(0.5, 0.8)$ and out-of-block probability 0.3.

Simulation 3: Network Model (Stochastic Block Model)



Simulation 3: Network Model (Stochastic Block Model)

Clustering ARI	$N = 20$				$N = 40$			
	JisstPCA (BIC)	G-JisstPCA (BIC)	iHOSVD (oracle)	iHOOI (oracle)	JisstPCA (BIC)	G-JisstPCA (BIC)	iHOSVD (oracle)	iHOOI (oracle)
Sample	0.947(0.238)	1 (0)	1 (0)	0.954(0.205)	1 (0)	1 (0)	1 (0)	1 (0)
Network 1 of $\mathcal{X}$	0.971(0.108)	1 (0)	0.912(0.195)	0.739(0.251)	1 (0)	1 (0)	0.805(0.241)	0.663(0.224)
Network 2 of $\mathcal{X}$	0.995(0.022)	0.997 (0.011)	0.146(0.057)	0.139(0.043)	1 (0)	1 (0)	0.156(0.015)	0.153(0)
Network 1 of $\mathcal{Y}$	0.974(0.114)	1 (0)	0.99(0.027)	0.94(0.155)	1 (0)	1 (0)	0.973(0.108)	1 (0)
Network 2 of $\mathcal{Y}$	0.87 (0.31)	0.867(0.309)	0.014(0.046)	0.121(0.16)	1 (0)	1 (0)	0.116(0.101)	0.246(0.153)
$\sin \theta(\hat{\mathbf{u}}_1, \mathbf{u}_1^*)$	0.087(0.03)	0.089(0.031)	0.138(0.051)	0.08 (0.026)	0.092(0.016)	0.093(0.017)	0.142(0.028)	0.083 (0.016)
$\sin \theta(\hat{\mathbf{u}}_2, \mathbf{u}_2^*)$	0.167(0.077)	0.163 (0.066)	0.214(0.04)	0.188(0.162)	0.152(0.014)	0.152(0.014)	0.197(0.017)	0.145 (0.015)
$\ \sin \Theta(\hat{\mathbf{V}}_1, \mathbf{V}_1^*)\ _{\text{op}}$	0.084 (0.007)	0.084 (0.007)	0.163(0.06)	0.163(0.059)	0.062 (0.006)	0.062 (0.007)	0.154(0.046)	0.156(0.047)
$\ \sin \Theta(\hat{\mathbf{V}}_2, \mathbf{V}_2^*)\ _{\text{op}}$	0.21 (0.074)	0.211(0.072)	0.817(0.065)	0.799(0.07)	0.154 (0.024)	0.155(0.023)	0.776(0.022)	0.768(0.028)
$\ \sin \Theta(\hat{\mathbf{W}}_1, \mathbf{W}_1^*)\ _{\text{op}}$	0.158 (0.018)	0.158 (0.018)	0.253(0.056)	0.196(0.058)	0.118 (0.014)	0.118 (0.014)	0.202(0.026)	0.173(0.04)
$\ \sin \Theta(\hat{\mathbf{W}}_2, \mathbf{W}_2^*)\ _{\text{op}}$	0.416 (0.204)	0.417(0.203)	0.969(0.044)	0.903(0.098)	0.272 (0.045)	0.273(0.045)	0.901(0.062)	0.771(0.079)

Empirical Results

Case Study: Integrating Multi-modal Population Brain Connectivity

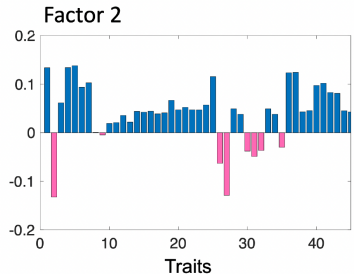
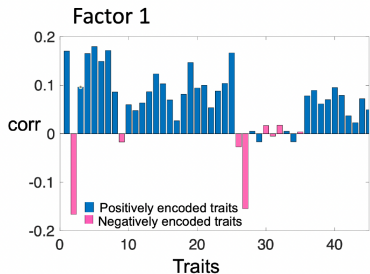
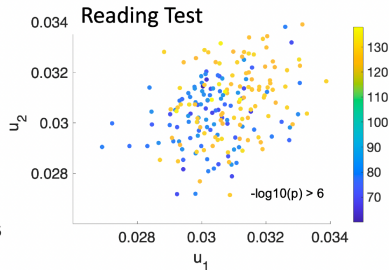
HCP Data & Preprocessing:

- 1065 subjects.
- 87 ROIs (68 cortical and 19 subcortical).
- Structural Connectome (SC): weighted edges based on log number of fiber curves between ROIs (Zhang et al., 2018).
- Functional Connectome (FC): Correlation between mean time series of ROIs (Glasser et al., 2013).
- Processed tensor modalities: $87 \times 87 \times 1065$.

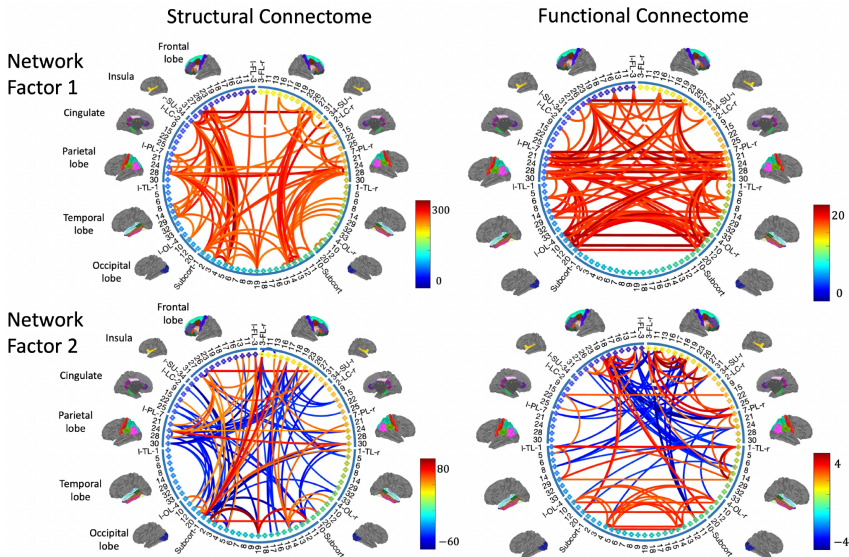
Our Analysis: Generalized JisstPCA

- $K = 2$ factors explained 85.5% of the variance in SC data and 77.9% in FC data.
- BIC Deflation chose rank 5 for both sets of factors.

Human Connectome Project Results



Human Connectome Project Results



Summary

- JisstPCA (G-JisstPCA) model well-suited to modeling populations of multi-modal networks.
- Power algorithm has one-step statistical convergence & achieves a comparable rate to that of Tucker models!
- Strong empirical results & interesting findings for multi-modal brain connectomes.
- **Future Work:** Multi-factor analysis, beyond PCA, and many applied directions on large-scale biobank data (e.g. integrating brain connectomes and genomics).

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Thank You!